

FINITENESS AND RIGIDITY: A TWO-LEMMA ARCHITECTURE FOR THE RIEMANN HYPOTHESIS

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ABSTRACT. Let $m \in \{0, 1, 2, \dots, \infty\}$ denote the number of quadruples of non-trivial zeros of ζ off the critical line, so that the Riemann Hypothesis (RH) is the statement $m = 0$. We prove a conditional dichotomy: if ζ admits pointwise self-approximation along a single unbounded sequence of shifts on compact disks of the open critical strip (Lemma LP-112, formally weaker than Bagchi's strong recurrence), then $m \in \{0, \infty\}$. The hypothesis $m < \infty$ freezes every constant in the Bohr–Rouché replication argument—a minimal off-line distance, a nonvanishing margin, a maximal height—so that a single replication suffices, and the Euler product enters at three identified points, excluding the known finite- m counterexamples without Euler product. Consequently RH follows from two lemmas: *(L8) finiteness*, $m < \infty$, which we argue is the countable avatar of cohomological finite-dimensionality over $\text{Spec } \mathbb{Z}$ (it admits no Siegel-type repulsion: two quadruples are *harder* to exclude than one, since index costs add rather than compete for a scarce budget); and *(LP-112) sequential rigidity*, for which we show there is no precedent in the literature: every known self-approximation theorem is a density statement proved by genericity, and the natural Montel-plus-Besicovitch chain dies at the identification of the limit. As a byproduct we prove an unconditional negative theorem in seminorm coordinates: ζ is not Stepanov almost-periodic, in the band (vector-valued) sense, on any sub-strip of the critical strip—every seminorm that sees compact windows is destroyed by the growth of ζ , and every seminorm under which ζ is almost periodic is blind to compact windows. The two lemmas reproduce the order of the only proof of an RH-type statement that exists (finiteness before purity, Weil–Deligne), and the wall of [10] is shown to be insensitive to the cardinality of witnesses demanded, sensitive only to their exceptionality. No proof of RH is claimed.

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1. INTRODUCTION

1.1. From a closed map to an open architecture. The companion paper [10] closed a long programme: it argued that every analytic, spectral, and combinatorial route to RH terminates at a single obstruction—the inversion of one quantifier, from averaged to individual—and that no argument-form catalogued there crosses it. A closed map invites a question it cannot itself answer: *if the wall is one, where is its gate?* This paper records the programme’s answer, found by asking a physicist’s question (what pins a system to its critical point, and what kind of object counts modes one by one) and following it to its mathematical end [7, Docs 109–113].

The answer is an architecture. Writing m for the number of off-critical zero quadruples,

$$\text{RH} = \underbrace{(m < \infty)}_{\text{finiteness}} \wedge \underbrace{(m < \infty \Rightarrow m = 0)}_{\text{dichotomy}},$$

and the two halves have different natures, different natural owners, and different walls. The dichotomy, we prove, follows from a single sequential rigidity lemma (LP-112) that is strictly weaker than every previously identified sufficient condition. The finiteness, we argue, is the countable avatar of cohomological finite-dimensionality, inaccessible to analysis for a structural reason (there is no Siegel-type repulsion among quadruples) and matching exactly what the Connes–Consani arithmetic-site programme would supply if it succeeds. This reproduces the order of the only existing proof of an RH-type theorem: over function fields, finiteness ($\dim H^1 = 2g$) came first, purity second.

We are explicit about status: both lemmas remain open, and both sit behind the master-quantifier wall of [10]—finiteness because m is an intensive, density-zero parameter invisible to averages; LP-112 because its witnesses form an exceptional set under $\neg\text{RH}$. The contribution is not progress along either lemma but the articulation itself: the joint at which the problem separates into a geometric half and an analytic half, with the boundary between them drawn precisely.

1.2. Framework and notation. We use the framework of [6, 7, 8, 9, 10]. Off-critical zeros of ζ come in quadruples $\{\beta \pm i\gamma, 1 - \beta \pm i\gamma\}$ with $\beta = \frac{1}{2} + \delta$, $\delta \in (0, \frac{1}{2})$; m counts the quadruples; $\text{RH} \iff m = 0$. By Bohr–Landau (1914), in every $\neg\text{RH}$ world the off-critical zeros have density zero among all zeros. The negative index of the Weil form satisfies $\kappa(Q) = 2m$ on the off-critical subspace $\mathcal{K}_{\text{off}}$ [5], [7,].

2. THE DICHOTOMY THEOREM

2.1. The rigidity lemma.

Definition 2.1 (LP-112: sequential self-approximation). For every compact disk $D \subset \{\frac{1}{2} < \text{Re}(s) < 1\}$ there exists a sequence $\tau_k \rightarrow \infty$ (which may depend on D) such that $\zeta(s + i\tau_k) \rightarrow \zeta(s)$ uniformly on D .

Remark 2.2 (Quantifier form). This is the *weak* (per-disk) form, and it is all the proof of Theorem 2.3 uses: a single disk around a single off-critical zero. The strong form (one sequence serving all disks simultaneously) is not needed and is not assumed.

LP-112 demands a *sequence* of good shifts—non-vacuity of the set of pointwise almost-periods—where Bagchi’s strong recurrence demands positive lower density of them. The gap in strength is strict and decisive: strong recurrence is *equivalent* to RH [7,], while LP-112 is implied by RH and, as far as every known dictionary reaches, does not imply it (§2.4).

2.2. The theorem.

Theorem 2.3 (Conditional dichotomy [7, Thm 2.3]). *Assume LP-112. Then $m \in \{0, \infty\}$. Consequently RH follows from LP-112 together with $m < \infty$.*

Sketch, with the role of finiteness made explicit. Suppose $0 < m < \infty$. Finiteness freezes every constant of the replication argument: there is a minimal off-line distance $\delta_{\min} > 0$, a maximal height $\gamma_{\max} < \infty$, and around a fixed off-critical zero ρ_0 a closed disk D of radius $r < \delta_{\min}/2$ on whose boundary $\eta := \min_{\partial D} |\zeta| > 0$. Take τ_k as in LP-112 with $\sup_D |\zeta(s + i\tau_k) - \zeta(s)| < \eta$ for large k . By Rouché, $\zeta(\cdot + i\tau_k)$ has a zero in D , i.e. ζ has a zero in $D + i\tau_k$ —off-critical, since $r < \delta_{\min}/2$ keeps the replica away from the line, and *new*, since $\tau_k \rightarrow \infty$ eventually exceeds $2\gamma_{\max}$. One replica gives $m + 1$ quadruples; iterating along the sequence gives infinitely many, contradicting $m < \infty$. Hence $m = 0$ or $m = \infty$. $\square$

Remark 2.4 (Why a single replica suffices, and why that matters). Every earlier replication argument in the programme (Bagchi’s, and its effective version [7,]) needed a positive-density family of replicas to contradict the zero-density theorems $N(\sigma, T) = o(T)$ —and was self-blocking precisely because the needed density and the available ceiling carried the same exponent. Under $m < \infty$ the target changes: *one* replica beats finiteness. The self-blocking mechanism of [7,] does not apply to the sequential version (no density theorem obstructs density-zero replication); what remains is only the exceptionality of the witnesses (§4.3).

2.3. Where the Euler product enters. By [7, Prop. 2.2], every finite m is realisable in the class of order-one real entire functions with the functional equation (explicit cosh-type examples), so Theorem 2.3 must use the Euler product essentially. It does, at three identified points [7, §3]: (i) the B^2 almost-periodicity that feeds any approximation statement comes from the Dirichlet series; (ii) the $\mathbb{Q}$ -linear independence of $\{\log p\}$ —unique factorisation, pure arithmetic—makes the group of almost-periods relatively dense and non-discrete, so replicas land at *new* heights (the cosh-examples are periodic with one frequency: their “replicas” land on the same zeros and generate nothing); (iii) the random Euler product underlies the support theorems that control the limit. The counterexamples are excluded at the first step: they are not even B^2 .

2.4. LP-112 is not known to be RH-equivalent. RH implies LP-112 (Bagchi’s theorem gives positive density of pointwise almost-periods, a fortiori a sequence). Conversely LP-112 yields only the dichotomy: under $\neg$ RH its consequence is $m = \infty$, which is consistent. LP-112 is formally weaker than strong recurrence (sequence versus positive density, a one-way implication), and [7,] verifies that no known dictionary upgrades it to RH. We state the status precisely: the non-equivalence is *not refuted*, not *proved*—indeed no material non-equivalence can be proved while RH is open, since under RH both statements are true. What is certified is that LP-112 is weaker *as a statement* along every known derivation; whether it is easier *as a problem* is the architecture’s declared bet, not a theorem. It is the weakest statement so far identified that, with finiteness, closes the argument.

3. THE FINITENESS LEMMA IS COHOMOLOGICAL

3.1. No Siegel mechanism. Could analysis prove $m < \infty$ directly—an unconditional “at most C exceptions” theorem, as for Siegel zeros? The analysis of [7,] says no, for a structural reason worth recording. Siegel-type repulsion works by *scarcity of a budget*: the simple pole of ζ is a global $O(1)$ resource near $s = 1$, and two exceptional zeros compete for it. For off-critical quadruples in the interior of the strip the local budget is the archimedean term $\asymp \log \gamma_0$, which *grows*; without scarcity there is no repulsion. Worse, the mechanism is *anti-Siegel*: in every counting functional of the programme the costs of two quadruples add—they never compete—so $m = 2$ is harder to exclude than $m = 1$. And the statistics that are unconditionally Gaussian-concentrated (Selberg’s central limit theorem for $S(t)$) are blind to the abscissa: a quadruple is indistinguishable from a double on-line zero for $N(T)$ and $S(T)$ [7, Lem. 5.1].

3.2. Finiteness as the countable avatar of $\dim H^1 < \infty$. The two historical precedents of an unconditional “finite-budget-per-object” theorem are Siegel’s (budget: the pole) and Weil’s over function fields (budget: $\dim H^1 = 2g$, established *before* purity). The programme’s Form B analysis sharpened this: the Deligne mechanism transplanted to (K, Q) possessed amplification but lacked precisely an a-priori finiteness source—“Form B was Deligne minus finiteness” [7,]—and the missing axiom (L6: a-priori finite rank with arithmetic traces) is exactly what the tropical Riemann–Roch of the Connes–Consani site does not yet provide: their theorem yields Euler characteristics (values), not signatures with positivity (inertia). Moreover, by [7, Prop. 4.5], the value/inertia wall of [10] is rank-sensitive: *in a-priori finite rank, finitely many autonomous traces determine the inertia* (Newton identities). Finiteness converts the unreadable into the readable. We therefore state the geometric half as a target specification rather than a conjecture of analysis:

Problem 3.1 (L8, the finiteness lemma). Prove $m < \infty$: the off-critical zeros of ζ , if any, are finite in number. Equivalently (in the programme’s operator form): the negative index $\kappa(Q) = 2m$ is finite a priori. The expected source is a cohomological finiteness theorem over $\text{Spec } \mathbb{Z}$ —the analogue of $\dim H^1 = 2g$ —not an analytic estimate; no analytic route catalogued in [7, 8, 9, 10] reaches it, and the anti-Siegel structure above explains why.

4. THE RIGIDITY LEMMA HAS NO PRECEDENT

4.1. The map of what is proved. [7,] surveys self-approximation exhaustively. Everything proved is a *density* statement established by *genericity*: the Hurwitz zeta self-approximates at $d = 0$ unconditionally because its orbit closure has full support (no Euler product—exactly the property ζ lacks); the family SR_d for $d \neq 0$ holds by equidistribution (Baker, six exponentials); $d = 0$ for ζ with positive density is Bagchi’s theorem, equivalent to RH—an equivalence already present, in almost-period language, in Bohr’s 1922 work, sixty years before Bagchi. The unconditional abscissa for pointwise almost-periodicity of ζ is exactly $\sigma_0 = 1$; the sequential statement is open for every $\sigma_0 < 1$. No class of functions is known in which sequential self-approximation is proved without either full support or an RH-equivalent.

4.2. A negative theorem in seminorm coordinates. The natural attack on LP-112—compactness (Montel) plus almost-periodicity (B^2)—fails at a precise and instructive point. The compactness links hold unconditionally: by the second-moment bounds, the set of τ with $\zeta(\cdot + i\tau)$ bounded on a fixed disk has density $1 - \delta$; intersecting with the positive-density set of B^2 almost-periods gives a sequence along which $\zeta(s + i\tau_k)$ converges to some holomorphic g . The chain dies at the *identification* $g = \zeta$: the B^2 seminorm is a global average in t and assigns weight zero to the compact window of the disk, so the certificate “ τ_k is a B^2 almost-period” carries no information about the window where the zero lives—the same locus as [7, Prop. 2.6]. Could a stronger seminorm see the window? We prove it cannot:

Theorem 4.1 (No Stepanov almost-periodicity, band sense [7, Thm 5.2]). *ζ is not Stepanov almost-periodic on any sub-strip of the open critical strip, in the band (vector-valued) sense: for every $[\sigma_1, \sigma_2] \subset (\frac{1}{2}, 1)$ and every $p \geq 1$, the vectorial Stepanov norm $\sup_x \int_x^{x+1} \int_{\sigma_1}^{\sigma_2} |\zeta(\sigma + it)|^p d\sigma dt$ is infinite, so $t \mapsto \zeta(\cdot + it)$ is not S^p almost periodic as an $L^p(\sigma_1, \sigma_2)$ -valued function. (Stepanov almost periodicity implies finiteness of the Stepanov norm, which the unconditional large-value theorems for ζ , via subharmonicity of $|\zeta|^p$ on disks contained in the band, violate.) The fixed-line scalar version is not claimed and remains open.*

Observation 4.2 (The wall in seminorm coordinates). Every seminorm that sees compact windows (uniform, Stepanov) is destroyed by the growth of ζ in the strip; every seminorm under which ζ is almost periodic (B^2 , Weyl) is blind to compact windows. The master quantifier of [10], in this coordinate system: the seminorms that average survive and certify nothing individual; the seminorms that individuate do not survive.

4.3. The wall, and a sharpening of [10]. Under $\neg\text{RH}$, the set of τ good for Rouché on the zero’s disk has upper density zero [7, Prop. 2.6]: LP-112 asks to certify that an exceptional set is non-empty. This is the master quantifier in sequential form, and it yields a sharpening of the no-go:

Observation 4.3 (Cardinality-insensitivity). The wall of [10] is insensitive to the cardinality of witnesses demanded—positive density of replicas or a single one—and sensitive only to their exceptionality. Weakening a target from “positive density” to “one witness” removes the self-blocking against density theorems [7,] but does not move the wall, because the witness still lives in a null set that averaging methods cannot enter.

The named missing piece is the anchoring lemma *LP-113* [7,]: localise the mean goodness of a B^2 almost-period into the fixed compact window. LP-113 implies LP-112, follows from RH, and is not RH-equivalent by any known dictionary—but its witnesses are exceptional under $\neg\text{RH}$: same wall, one level down.

5. THE ARCHITECTURE, ASSEMBLED

Theorem 5.1 (Two-lemma reduction). *RH follows from L8 (Problem 3.1) together with LP-112 (Definition 2.1). The two lemmas are independent in nature: L8 is a finiteness statement of cohomological type, with no analytic route and no Siegel mechanism; LP-112 is an analytic rigidity statement, strictly weaker than every previously identified sufficient condition, with no precedent in any function class. Both sit behind the master-quantifier wall of [10]: L8 because m is intensive and density-zero, LP-112 because its witnesses are exceptional.*

Remark 5.2 (The Weil order). The architecture reproduces the structure of the one proof that exists. Over function fields, finiteness ($\dim H^1 = 2g$, cohomological) preceded purity (the zeros where they must be); and given finiteness, finitely many traces—Lefschetz numbers, the autonomous values—determine everything. Here likewise: given L8, the rank is finite a priori, finitely many arithmetic traces determine the inertia [7, Prop. 4.5], and the replication argument of Theorem 2.3 needs only one witness. The programme spent thirty-seven phases attempting purity without finiteness; the architecture explains why that order was impossible.

Remark 5.3 (The physics that found the joint). The articulation was located by physical reasoning [7,]: the Rodgers–Tao theorem ($\Lambda \geq 0$) is one-sided self-organised criticality, provable because its order parameter (crystalline rigidity) is *extensive*—visible to averages; the order parameter of the other side is m itself, *intensive* and density-zero by Bohr–Landau, so no relaxation sandwich exists. The thermodynamic reading of the master quantifier (extensive = average = theorem; intensive = individual = wall) is what suggested splitting RH at $m < \infty$, where the intensive parameter becomes a priori finite and the wall becomes, in principle, crossable from the other side.

6. HONEST ASSESSMENT

What is proved here and in the underlying documents: the conditional dichotomy (Theorem 2.3); the three essential entry points of the Euler product, with the finite- m counterexamples excluded; the non-equivalence of LP-112 to RH by all known dictionaries; the absence of any precedent for sequential self-approximation without genericity; the unconditional Stepanov negative theorem (Theorem 4.1) and the seminorm dichotomy (Observation 4.2); the cardinality-insensitivity of the wall (Observation 4.3); and the anti-Siegel structure of the finiteness problem.

What is not proved: either lemma. We do not predict that either will be proved. The honest summary is this: the programme that [10] closed has, in closing, located the gate it could not open. RH separates at its natural joint into a geometric finiteness statement—whose specification matches what the arithmetic-site programme is building toward, and which we hand to geometry—and an analytic rigidity statement weaker than anything previously known to suffice, whose exact obstruction is now a theorem about seminorms rather than a fog. If the gate opens, it opens from

the geometric side first, exactly as it did the only other time. Mapping a wall is not crossing it; but a map that ends at a gate, with the gate's two locks named and specified, is the most a programme of this kind can honestly leave behind.

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